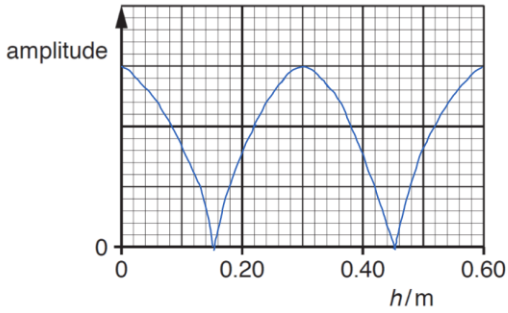
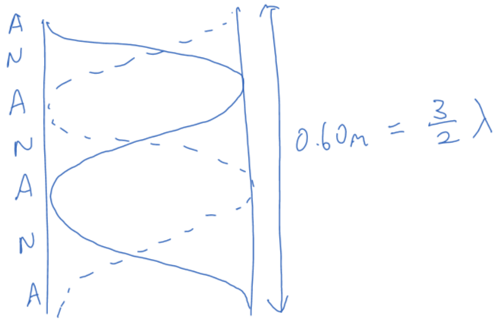
5ai	The incident wave reflects at the top of the tube. The incident and reflected wave interfere / superpose to form the stationary wave.	**A**	B1 B1
5aii	 Maximum value at *h* = 0, 0.30 m, 0.60 m, and 0 amplitude at *h* = 0.15 m, 0.45m. The line should be a <u>modulus</u> of a sinusoidal function.	**D**	B1 B1
5aiii1	vertical/along length of tube/along axis of tube	**E**	B1
5aiii2	phase difference = 180°	**E**	B1
5aiv	When next stationary wave is formed: 0.60 m = 1.5 λ v = f λ, so f = v/λ = 340 / [0.60/1.5] = 850 Hz 	**A**	C1 A1
5bi	The waves <u>spread</u> as they pass through the <u>slits</u>.	**A**	B1

5bii	$d \sin \theta = n\lambda$ $\sin \theta = n\lambda/d$ since $\sin \theta \leq 1$, $n\lambda/d \leq 1$ <u>when n=3 and $\lambda = 540 \times 10^{-9}$ m</u> $3(540 \times 10^{-9})/d \leq 1$ $d \geq 1.62 \times 10^{-6}$ m <u>when n=3 and $\lambda = 630 \times 10^{-9}$ m</u> $3(630 \times 10^{-9})/d \leq 1$ $d \geq 1.89 \times 10^{-6}$ m to satisfy both conditions above, minimum d is 1.89×10^{-6} m	**D**	C1 A1
5biii	<u>wavelength</u> of blue light is <u>shorter</u> (than both given lights) so the third order diffraction maximum is produced.	**A**	B1

6ai	arrow from –0.85 eV level to –1.50 eV level	**E**	B1